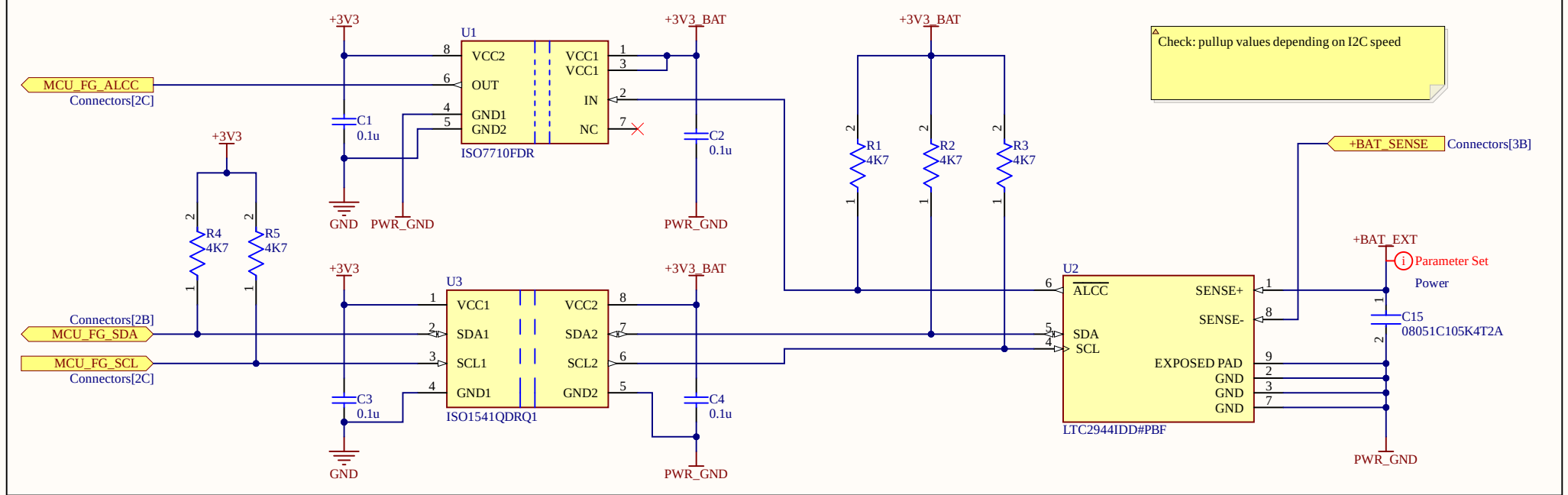
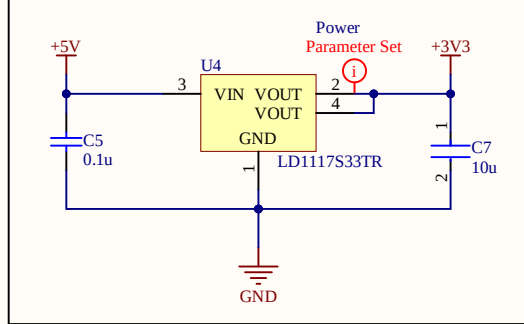


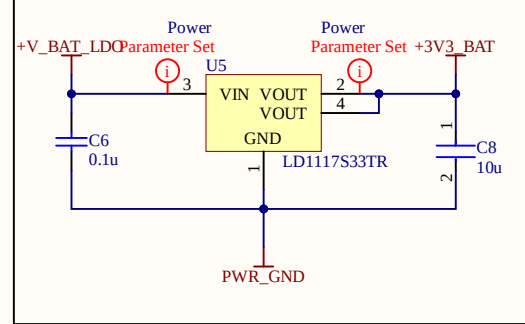
Battery Fuel Gauge



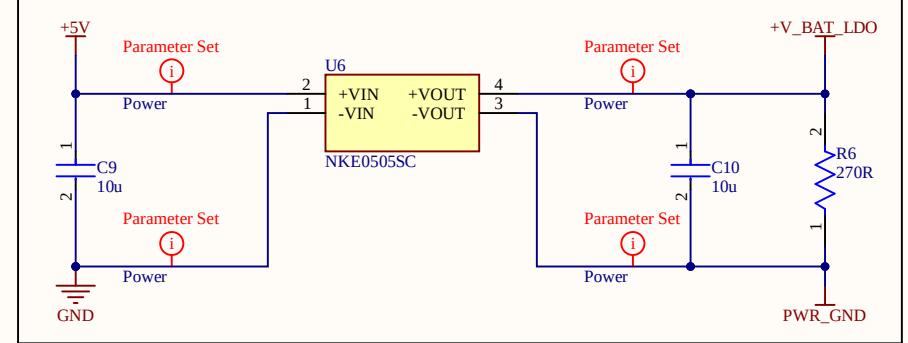
+3V3 MCU Side LDO



+3V3 Battery Side LDO

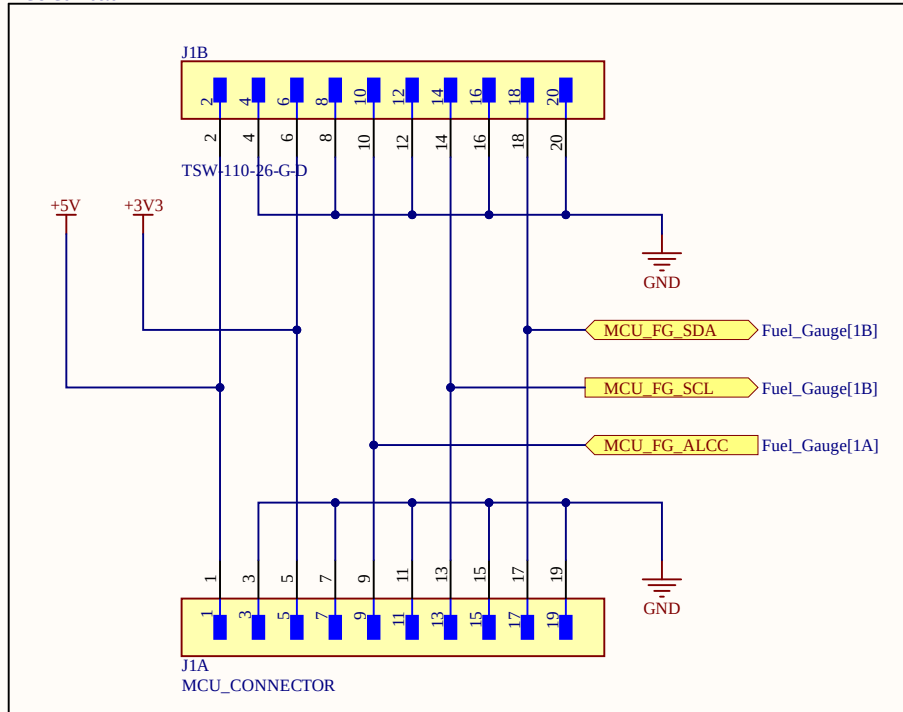


+5V Isolated

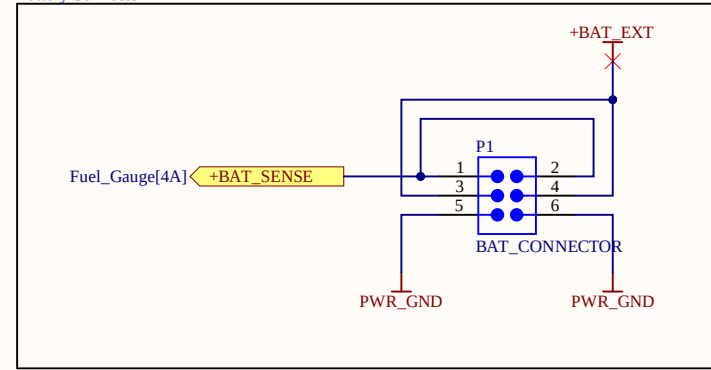


Title		
Size	Number	Revision
A4		
Date:	8/27/2026	Sheet of
File:	C:\Users\...\Fuel_Gauge.SchDoc	Drawn By:

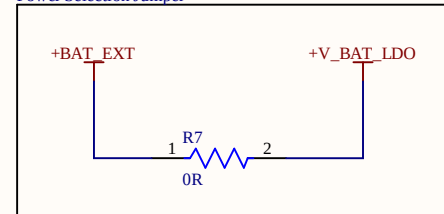
MCU Connector



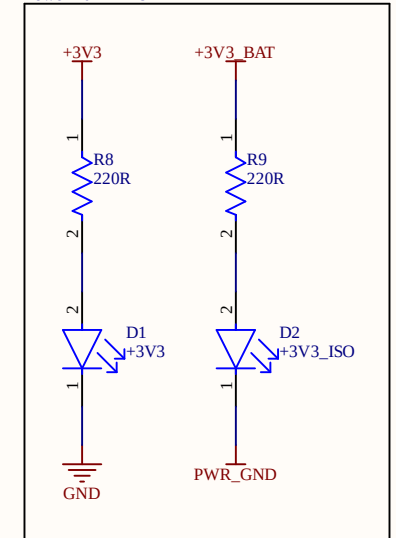
Battery Connector



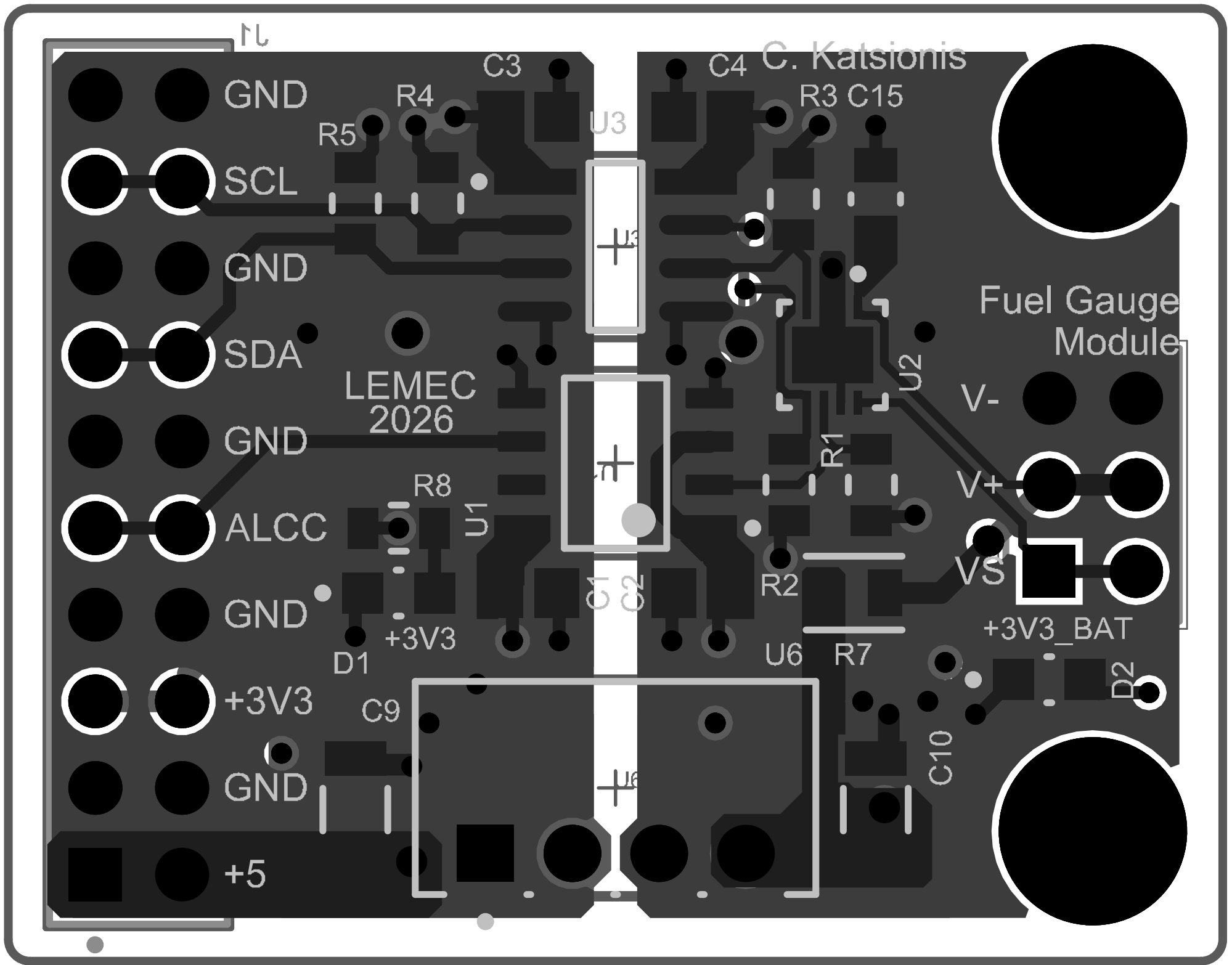
Power Selection Jumper



Power Rail LEDs



Title		
Size	Number	Revision
A4		
Date:	8/27/2026	Sheet of
File:	C:\Users\...\Connectors.SchDoc	Drawn By:



IL

GND

SCL

GND

SDA

GND

ALCC

GND

+3V3

GND

+5

LEMEC
2026

R4

R5

C3

U3

U3

U3

U3

U6

C4

C. Katsionis

R3

C15

U2

R1

R2

U6

R7

U2

V-

V+

VS

+3V3_BAT

D2

C10

D2

D2

D2

Fuel Gauge
Module

Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	0.1u		C1, C2, C3, C4, C5, C6		Unknown server		6	Vishay	VD805Y104KXAMR	Volume Production	Arrow Electronics	VD805Y104KXAMR	0,0624	0,3744
	10u	CAP CER 10UF 50V X5R 1206	C7, C8, C9, C10		Unknown server		4	Yageo Group	CC1206K05F09BB106	Volume Production	Newark	CC1206K05F09BB106	0,467	1,87
	08051C105K4T2A	08051C105K4T2A Cap Ceramic 1uF 100V X7R 10% Pad SMD 0805 125C Automotive T/R	C15		Unknown server		1	KYOCERA AVX	08051C105K4T2A	Volume Production	Mouser	581-08051C105K4T2A	0,36	0,36
	+3V3, +3V3, ISO	LED GREEN CLEAR SMD	D1, D2		Unknown server		2	Diodes Inc.	LTST-C170GKT	Volume Production	DigiKey	160-1179-1-ND	0,17	0,34
	MCU_CONNECTOR	CONN HEADER VERT 20POS 2.54MM	J1		Unknown server		1	Samtec	TSW-110-26-G-D	Volume Production	Amet	TSW-110-26-G-D	2,53	2,53
	BAT_CONNECTOR	0.025" SQ Post Header, Through-hole, Vertical, -55 to 125 degC, 2.54 mm Pitch, 6-Pin, Male, RoHS	P1		Unknown server		1	Samtec	TSW-103-08-G-D	Volume Production	Amet	TSW-103-08-G-D	0,6732	0,6732
	4K7	RT0805 4.7KQ 0.125W 50 ppm/°C 0.01%	R1, R2, R3, R4, R5		Unknown server		5	Yageo Group	RT0805FRE134K7L	Volume Production	Mouser	603-RT0805FRE134K7L	0,017	0,17
	270R	Anti-Sulfurated Chip Resistor Automotive Grade 270Q 5% 0.25W ±100ppm/°C 1206	R6		Unknown server		1	Yageo Group	AF1206JR-07270RL	Volume Production				
	0R	Anti-Sulfurated Chip Resistor Automotive Grade 0Q Jumper 0.125W 0805	R7		Unknown server		1	Yageo Group	AF0805JR-070RL	Volume Production	DigiKey	13-AF0805JR-070RLCT-ND	0,1	0,1
	220R	RES SMD 220 OHM 1% 1/8W 0805	R8, R9		Unknown server		2	Yageo Group	RT0805FRE07220RL	Volume Production	Mouser	603-RT0805FRE07220RL	0,017	0,17
	ISO7710FDR	DGTL ISO 3000VRMS 1CH GP 8SOIC	U1		Unknown server		1	Texas Instruments	ISO7710FDR	Volume Production	Arrow Electronics	ISO7710FDR	1,16	1,16
	LTC2944IDD#PBF	IC BATTERY GAS GAUGE I2C 8DRN	U2		Unknown server		1	Analog Devices	LTC2944IDD#PBF	Volume Production	Newark	LTC2944IDD#PBF	10,43	10,43
	ISO1541QDRQ1	AUTO ISO I2C 2500VRMS 2CH 8SOIC	U3		Unknown server		1	Texas Instruments	ISO1541QDRQ1	Volume Production	Mouser	595-ISO1541QDRQ1	6,02	6,02
	LD1117S33TR	Fixed Low Drop Positive Voltage Regulator, 3.3V, 4-Pin SOT-223, Tape and Reel	U4, U5		Unknown server		2	STMicroelectronics	LD1117S33TR	Volume Production	DigiKey	497-1242-6-ND	0,42	0,84
	NKE0505SC	DC DC CONVERTER 12V 1W	U6		Unknown server		1	Murata Power Solutions	NKE1212SC	Volume Production	Mouser	580-NKE1212SC	7,66	7,66